

MLOps Assignment 2

DistilBERT-based Goodreads Genre Classifier

IIT Jodhpur | PGD in AI Programme

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1. Model Selection Rationale

We chose distilbert-base-cased for this assignment. DistilBERT is a distilled version of BERT that retains 97% of BERT's language understanding while being 40% smaller and 60% faster at inference. The cased variant preserves capitalization, which is useful for distinguishing proper nouns in book reviews (e.g., character names, titles). For a 7-class genre classification task on relatively short review texts, DistilBERT offers an excellent balance between computational cost (fits within Kaggle's free GPU quota) and classification performance. Its smaller size also means faster training iterations, allowing more experimentation within limited GPU hours.

2. Kaggle Training Setup

Platform Configuration:

- Accelerator: GPU T4 x2 (enabled under Settings → Accelerator)
- Internet: Enabled (required to download data, push to HF Hub)
- Secrets configured via Add-ons → Secrets:
 - WANDB_API_KEY — from wandb.ai/settings
 - HF_TOKEN — from huggingface.co/settings/tokens
- Runtime: ~15 minutes for 3 epochs on 5600 training samples

Training Hyperparameters:

Hyperparameter	Value
Epochs	3
Train Batch Size	16
Eval Batch Size	32
Learning Rate	3e-5
Warmup Steps	100
Weight Decay	0.01
Max Sequence Length	512

3. Evaluation results

Genre	precision	recall	f1-score	support
comics_graphic	0.83	0.79	0.81	200
fantasy_paranormal	0.48	0.48	0.48	200
history_biography	0.68	0.62	0.65	200
mystery_thriller_crime	0.54	0.60	0.57	200
poetry	0.78	0.83	0.80	200
romance	0.65	0.55	0.60	200
young_adult	0.32	0.34	0.33	200
accuracy			0.60	1400
macro avg	0.61	0.60	0.61	1400
weighted avg	0.61	0.60	0.61	1400

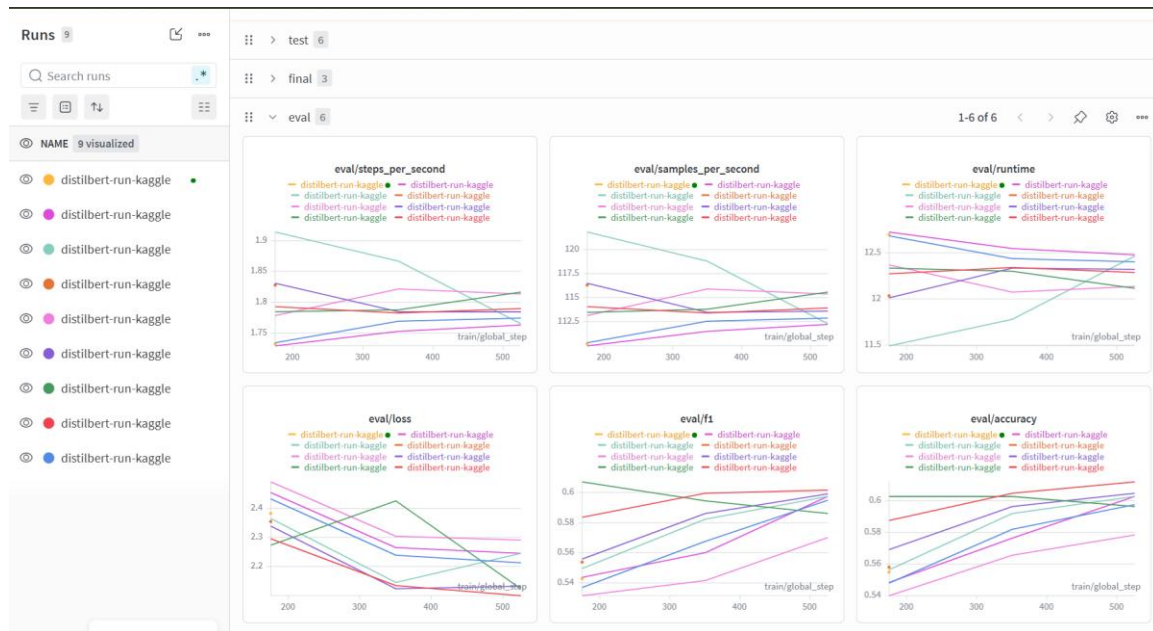
3. Results & W&B Dashboard

Final evaluation metrics on the held-out test set :

Metric	Score
Accuracy	0.60214
F1 Score (Weighted)	0.60667
Eval Loss	1.0753

W&B automatically captured training loss (every 50 steps), validation loss, accuracy, and F1 (each epoch), learning rate schedule, and GPU utilization. The eval_report.json was uploaded as a versioned W&B Artifact for reproducibility.

W&B Dashboard Screenshot:



4. Project Links

Resource	URL
GitHub	https://github.com/g25ait2033-cyber/saiharshith
Kaggle Notebook	https://www.kaggle.com/code/g25ait2033/mlops-assignment-2
W&B Dashboard	https://wandb.ai/g25ait2033-prom-iit-rajasthan/saiharshithmlops-2
Hugging Face Model	https://huggingface.co/g25ait2033-cyber/distilbert-goodreads-genres

5. Challenges & Learnings

- Dataset download: The UCSD Goodreads URLs occasionally timed out on Kaggle. Solved by implementing pickle-based caching so re-runs skip the download step.
- W&B run continuity: After training finishes, the W&B run closes. Logging evaluation metrics required resuming the same run using `wandb.init(id=..., resume="must")`.
- Hugging Face push from Kaggle: Initially failed because Internet was toggled off. Enabling Internet under Session Options resolved this.
- Genre confusion: Poetry and Young Adult reviews often overlap in language style, contributing to lower per-class F1 for these genres.

If I were to redo this assignment, I would experiment with larger sample sizes per genre and try learning rate scheduling with a cosine decay to potentially improve convergence.